In the near future, robots will handle an increasing share of everyday physical tasks. We examined, firstly, the comparative suitability of humanoids vs custom robots for tasks in households, warehouses, and factories, and how that difference in economic viability might change over time. And second, we analyzed the unit economics of a physical human job that will fall among the first to be fully automated house construction.

1. **Humanoids or task-specific machines: which wins in homes, warehouses and factories, and why? On what timeline will custom robotics make humanoids uneconomical?**

We focused mainly on robotics in households. This is the market that affects the most people, and moreover, both task-specific and humanoid robots have a chance of seizing market share. By contrast, robots in warehouses and factories only need to do one task (or a few similar tasks) millions of times. Task-specific robots that lack human form are much better suited for these specialized tasks, and already have a head start over humanoids in those environments.

The process:

1. We decided to simulate what a menagerie of task-specific robots would look like. This, of course, is only our best guess, and it's very much possible that further itemization may occur
2. We then compared: one humanoid that does all household tasks in all situations, vs several machines, each specialized for one task
3. We estimated the cost and the amount of learning that would need to be done - this tells us how quickly these systems can hit the market, and how affordable they could be.

Inside of our GitHub, you will see our simulation for what a house automated with task-specific robots looks like, based off of our extrapolations. Below, we will analyze the cost effectiveness of each system we implemented.

Results:
Robots best suited to take over household work must become available to consumers at an earlier timescale, and for a lower cost, than their competitors. Thus, to compare the viability of humanoid vs task-specific robots, we considered manufacturing costs and training time.
Our initial expectations: task-specific robots eliminate the need for manufacturers to buy certain redundant components, and they can each be trained for just one task. The disadvantage is that humanoids have a headstart in household work - any task-specific robot would have to make up for that by offering affordability or an accelerated training timeframe.
We excluded installation costs for certain systems, such as the bedmaker and the ChefRobot. This is because the cost may vary depending on the house. We would also need to design specific business models around such things. For example, would the bed be installed in someone's bed, or would a company construct a bed with robotic arms to already conduct all the tasks?

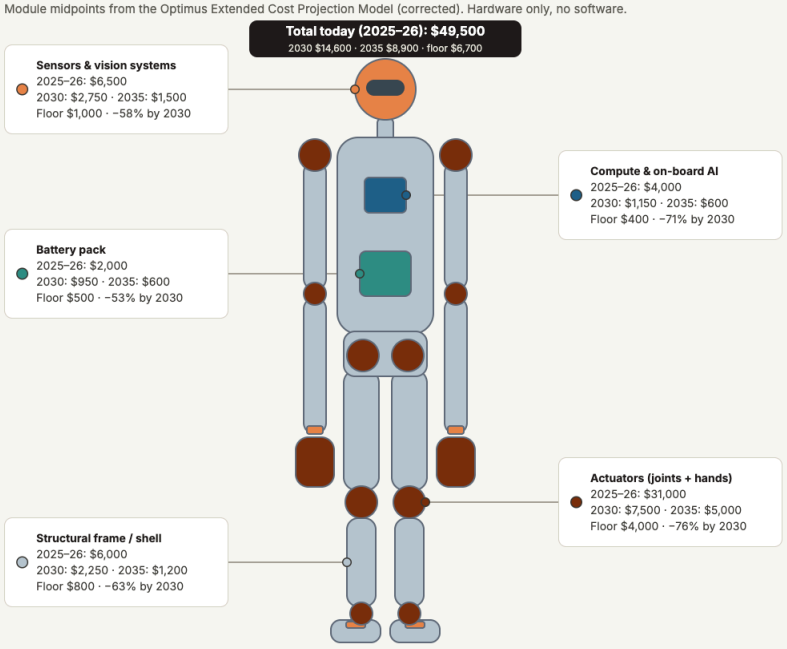
Robots that must work in less predictable environments with more sensor noise must require more training time. At one extreme, robots that must fold clothes or handle food will have to

process high-dimensional data with a lot of physical dexterity, thus requiring about as much training as simply training a humanoid for general tasks. At the other end, a trash bot simply has to move around, tip out old trash bags, and put on new ones, which requires significantly less training. By similar reasoning, we estimate that the bed and cleaning robots will each require more training than the trash bot and less training than a laundry or chef robot.

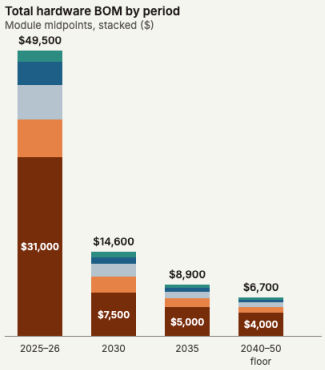
To estimate economic costs for each type of robot, we first referenced estimates of the cost of each component of the tesla optimus, to estimate how the costs of every component of a humanoid robot would change over the next few decades. We used projections from Morgan Stanley, Bain, and the Bank of America, to estimate the costs of key components.

Optimus hardware cost, 2025 to 2050

Module midpoints from the Optimus Extended Cost Projection Model (corrected). Hardware only, no software.



Module locations on the figure are indicative. The spreadsheet gives system-level costs, not costs per body part.

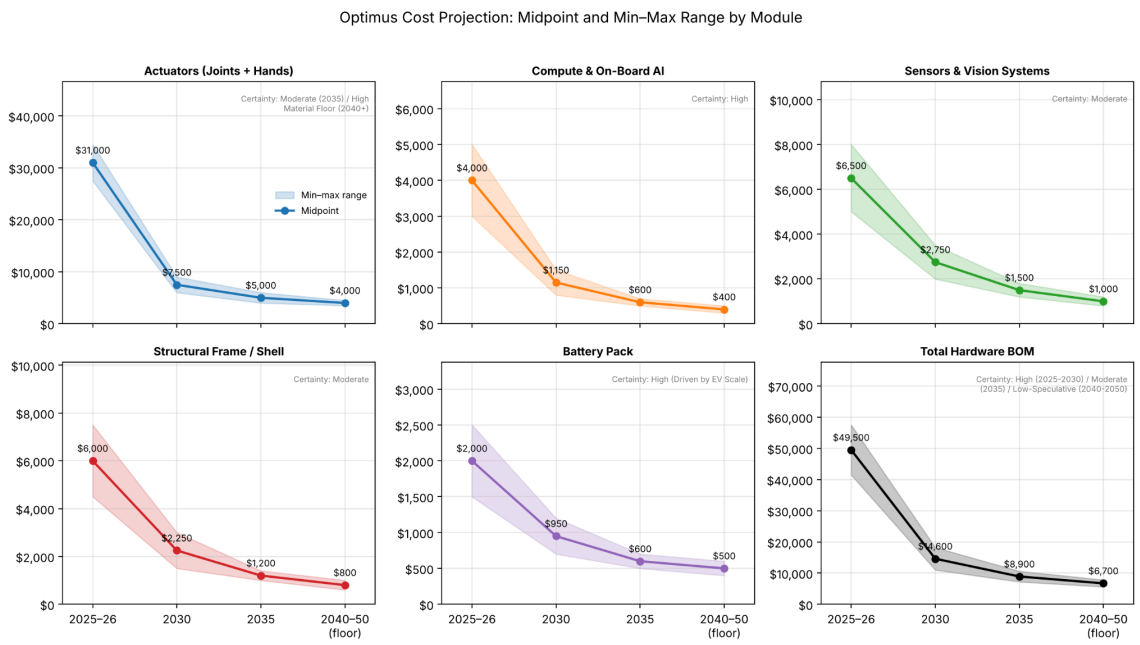


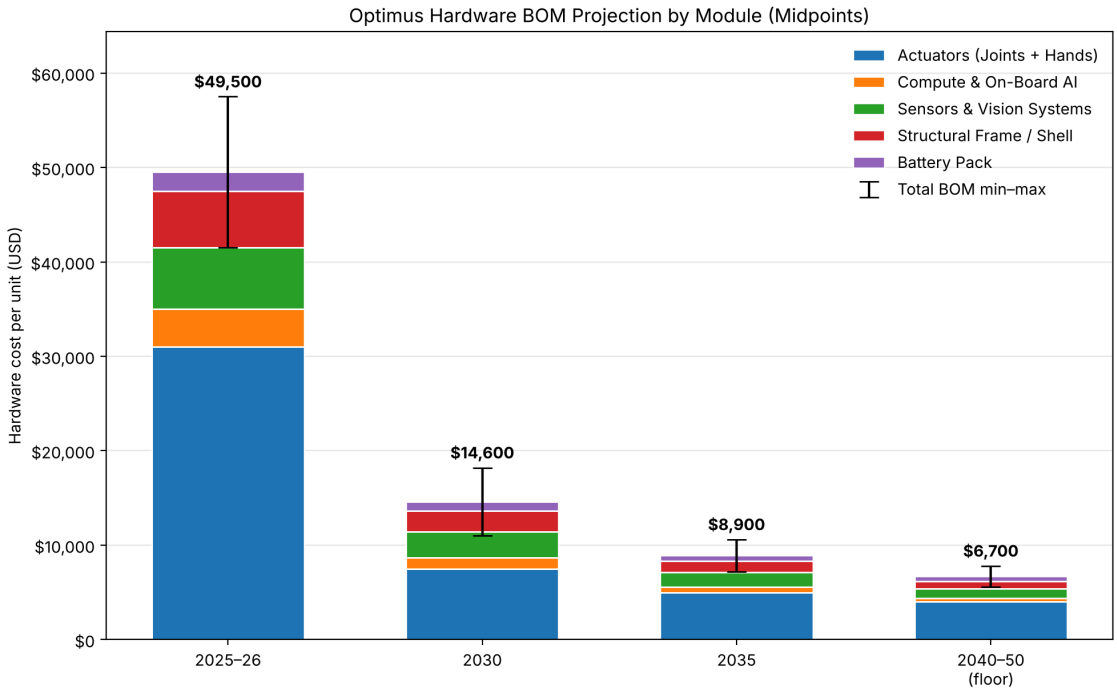
Flags from the spreadsheet's own notes

- Battery midpoint ($2,000) is about 6× Morgan Stanley's Gen 2 teardown ($0.3k).
- The 2035 total ($8,900) and the floor ($6,700) sit below every published forecast it lists.
- Actuators are 51% of the 2030 total, against about 70% in BofA's 2030 mix (incl. hands).
- Most values are model estimates; source checks used secondary coverage.

Module midpoints, with min–max range on the second line

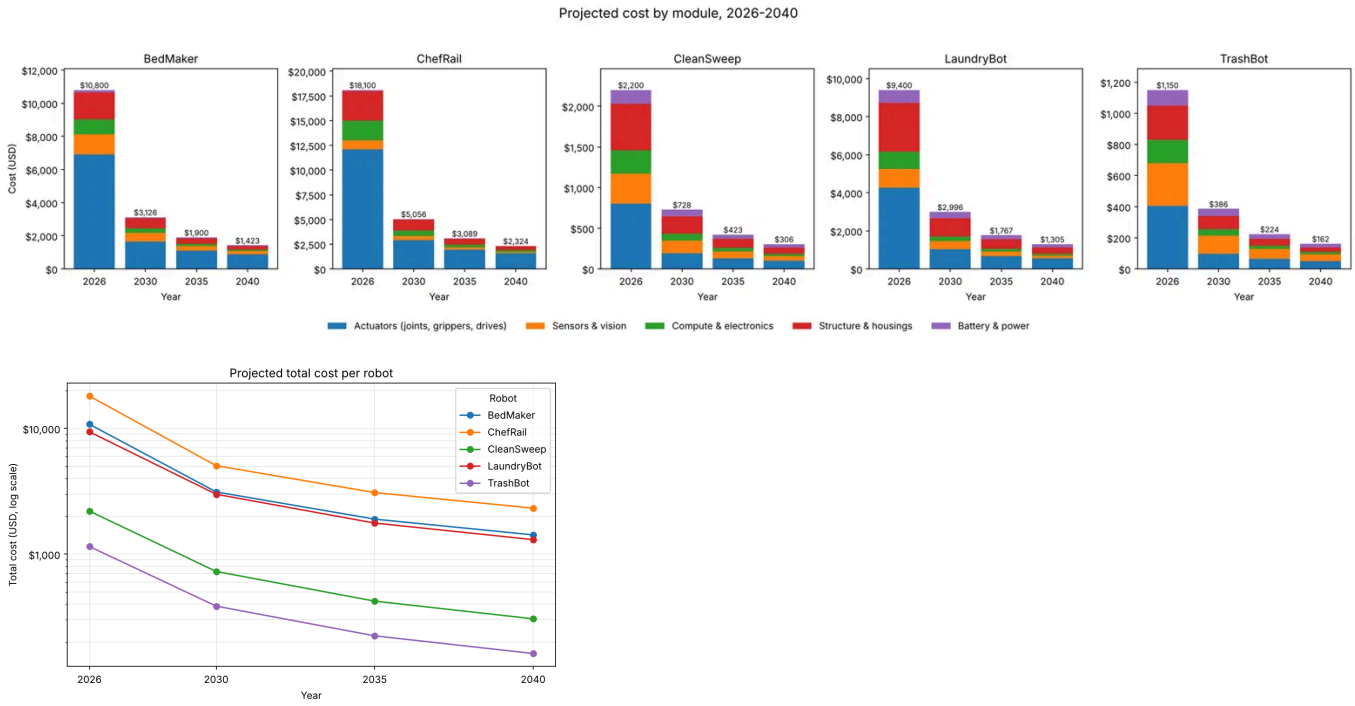
Module / system	2025–26 midpoint	2030 midpoint	2035 midpoint	2040–50 floor	2030 vs 2025–26	Long-term horizon certainty
Actuators (Joints + Hands)	**$31,000** 27,500–34,500	**$7,500** 6,000–9,000	**$5,000** 4,000–6,000	**$4,000** 3,500–4,500	**−75.8%**	Moderate (2035) / High Material Floor (2040+)
Compute & On-Board AI	**$4,000** 3,000–5,000	**$1,150** 800–1,500	**$600** 500–700	**$400** 300–500	**−71.2%**	High
Sensors & Vision Systems	**$6,500** 5,000–8,000	**$2,750** 2,000–3,500	**$1,500** 1,200–1,800	**$1,000** 800–1,200	**−57.7%**	Moderate
Structural Frame / Shell	**$6,000** 4,500–7,500	**$2,250** 1,500–3,000	**$1,200** 1,000–1,400	**$800** 600–1,000	**−62.5%**	Moderate
Battery Pack	**$2,000** 1,500–2,500	**$950** 700–1,200	**$600** 500–700	**$500** 400–600	**−52.5%**	High (Driven by EV Scale)
Total hardware BOM	**$49,500** 41,500–57,500	**$14,600** 11,000–18,200	**$8,900** 7,200–10,600	**$6,700** 5,600–7,800	**−70.5%**	High (2025-2030) / Moderate (2035) / Low-Speculative (2040-2050)





2035 and 2040–50 totals sit below published analyst forecasts and are speculative extrapolations.

Based on the costs of these components, we then determined cost estimates for the task-specific machines, which consisted of some common parts, but without redundant components (for instance, a specialized robotic chef would not need a leg-like locomotion system).



Based on these cost estimates, we note that while all the specialized robots are individually cheaper than a humanoid, the outsized costs of the BedMaker, LaundryBot, and ChefRail mean that in combination, all 5 task-specific robots would cost close to the same as one humanoid.

That said, certain specialists like the trashbot and cleansweep have a significant advantage over humanoids - they cost orders of magnitude less, and will likely take less training time, allowing them to hit the market earlier and at much lower costs.

Cost-wise, the discrepancy between humanoid and task-specific robots remains present over time. Humanoids are more dependent on actuators, and custom robots rely more on other components like batteries, so an economic shock to inputs like lithium or neodymium might shake up the balance of power. But this isn't a huge enough difference to grant one type of robot an edge in the fight for market share. The deciding factor regarding when (if ever) task-specific robots take over from humanoid generalists is how quickly each type of robot can be trained and made market-ready, and how quickly manufacturers of each type of robot can implement cost-saving design optimizations.

We conclude that, as both first-movers and an operational fit for repeating identical or similar tasks, task-specific robots have a clear edge over humanoids when it comes to factories and warehouses. However, household tasks will likely be split between general-purpose humanoids handling tasks that require dexterity or robustness to noise, and more specialized, task-specific robots for simpler, repetitive tasks. And since humanoids are more established than custom robots in the sphere of household tasks, they will likely hold greater market share for up to five years, until more affordable specialists become widely available to consumers. Such an

inflection point can come earlier or later, depending on different robots' relative time until market readiness.

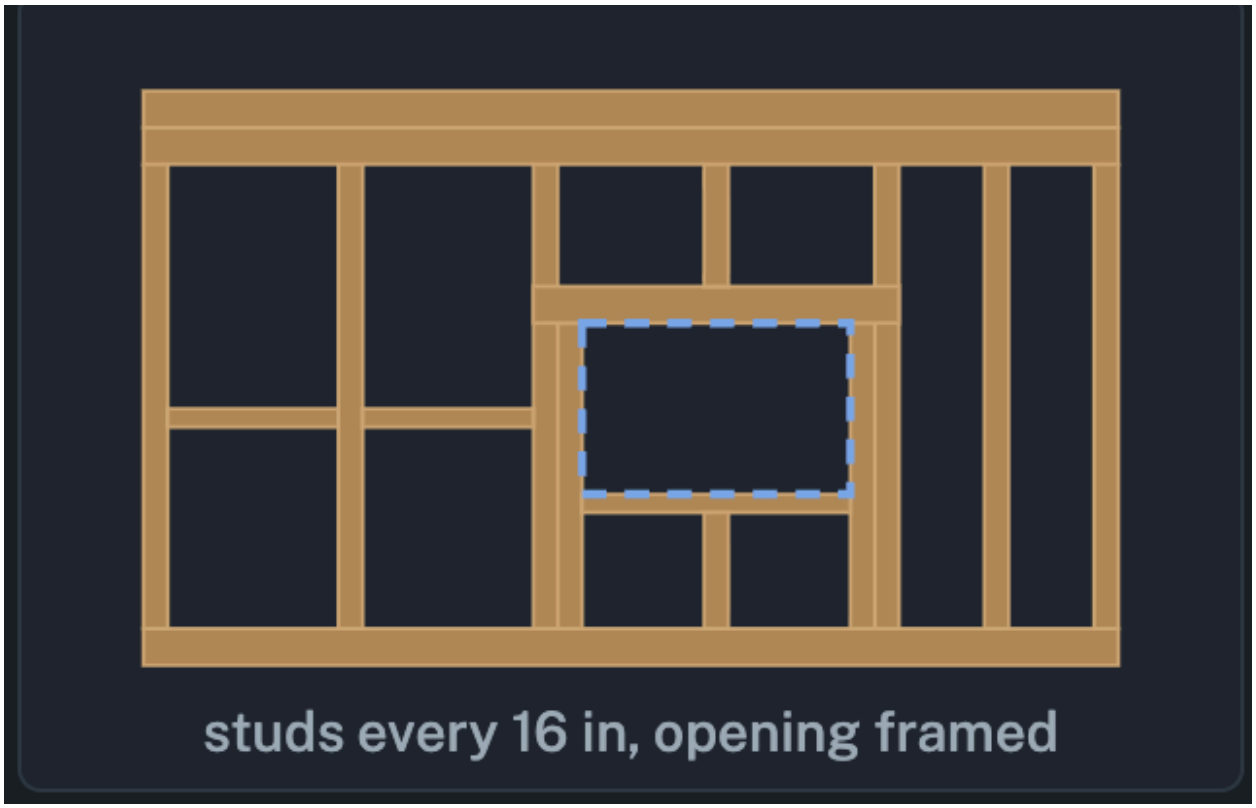
2. **Which job will robots take over first at scale, and what do the unit economics look like against human labor?**

To approach this question, we first focused on what kinds of tasks in the workforce can be automated. We concluded that jobs capable of being replaced by robots share specific characteristics: they have high precision and consistency, follow repetitive and structured procedures, lack a necessity for human connection, or involve dangerous and physically demanding conditions. To narrow our focus, we are moving past areas where robots are already widely implemented, like supermarkets, restaurants, and agriculture, and instead exploring sectors where robotics are less common. Based on these initial criteria, we determined that construction jobs are prime candidates for future automation. Construction emerges as a target for robotics because much of the labor involves repetitive, heavy-duty tasks that don't need human-to-human interaction such as bricklaying and material hauling. Construction is also a physically demanding job that can put human workers at risk in hazardous conditions.

Currently, modular housebuilding is a small yet rapidly growing sector in the United States. In modular housebuilding, instead of constructing every part of a home on-site, construction workers instead build large chunks of various houses at scale in warehouses. This process is still very much manual. In order to speed up the system, robotics can be implemented in various parts of the construction process. Let's take the construction of panels as an example.



Panels are constructed from strips of wood nailed together. The sawing process is only partially automated, since difficult or warped pieces of wood oftentimes compel manual involvement. Training a robotic system on wood-cutting could take the human out of the equation.



After the wood is fully cut out, each piece is nailed together into a frame which will form a part of a house's wall. The current system involves placing the pieces on a table, and nailing each together one by one. By dividing this step into two parts: one in which a robot arranges the wooden strips according to the software layout, and the other in which a robot handles the repetitive nailing process.



Placing a sheath over the completed board is not a difficult process, and may not even require advanced robotics. Many modular factories already automate the sheathing process.

Ultimately, these repetitive steps, automated away, leave only the delicate electrical and water piping steps, as well as final inspection, up to humans. Eventually, these roles may also be automated away by humanoid robots, though as such tasks are non-repetitive and require independent judgement, such roles are unlikely to be replaced anytime soon. However, the general construction work involved in such projects are likely to be automated away. As modular homes are already more cost efficient than regular homebuilding, as we will further explain below, regular construction work, which thus far has been insulated from the AI boom, may become indirectly automated away.

To understand whether robots could make home building cheaper, we built a cost model for a typical one-story house in the Bay Area. The house has the same features as the one we simulated in the previous question. It has 1,700 square feet of living space. We compared two approaches: traditional on-site construction with human crews, and a modular approach where robots build most of the house inside a factory before it is shipped to the site and assembled. The numbers below are planning estimates based on our own assumptions, not quotes from builders, due to the lack of existing data and the fact that real costs can vary widely by city, design, and company.

For a house like ours built the traditional way, we estimate labor at about $500,000. This does not include the cost of materials. We got this by estimating the labor hours and multiplying that by typical Bay Area billed rates. Our results show about 3,700 hours of labor, 30 to 50 different workers, and 7 to 12 months of construction, which leaves us with a range of $340,000 to $735,000, so we settled on a middle ground of $500,000.

We also estimated what it would cost if the modular process were fully automated, with robots doing all the construction work both inside the warehouse and on the building site. We estimate about $20 million in factory equipment and about $8.9 million a year to run it, or roughly $94,000 per house at 100 houses a year. Three on-site robots would cost about $10 million and about $5.1 million a year, or roughly $51,000 per house. With about $30,000 for transport, the total comes to roughly $175,000 per house. That number rises to about $235,000 once we include the estimated $25 million in research and development needed to build and train robots that don't exist yet. The approach only beats traditional building at about 33 to 48 houses a year, and the full upfront investment could reach $55 to $95 million. However, even a highly automated build would still need people, because inspections, utility hookups, truck driving, and robot maintenance are required. All of these figures are our own planning estimates based on industry knowledge and assumptions, not quotes from builders or robot makers.

To conclude, when we compare unit economics, the answer depends almost entirely on volume. For a single 1,700-square-foot house in the Bay Area, human labor costs about $500,000 in our estimate and a fully automated modular approach to construction could bring it to roughly $175,000, or about $235,000 once research and development are included. The savings of the modular construction depend on keeping the factory busy. Robots and factories have large fixed costs that are paid whether they build ten houses or a hundred. Below about 30 to 50 houses a year, automation costs more per house than human crews, but above that point, automation becomes cheaper than human labor by a wide and growing margin.

The most realistic path is a hybrid: a factory where robots handle repetitive, precise work like framing, drywall, insulation, and painting, while skilled workers handle the complex, judgment-heavy tasks. As robots improve and demand for housing grows, that balance can shift step by step toward automation. In a region with expensive labor and a shortage of construction workers like the Bay Area, that shift could make homes faster and cheaper to build.